

Deep Learning Foundations

Chapter 10: Convolutional Neural Networks

Sashreek Kumar

10.3)

Solution:

Calculating each element of the output matrix Y by applying the kernel to the input patches:

$$Y_{1,1} = (-2 \times 2) + (0 \times 5) + (4 \times 0) + (6 \times 6) = 32$$

$$Y_{1,2} = (-2 \times 5) + (0 \times -3) + (4 \times 6) + (6 \times 0) = 14$$

$$Y_{1,3} = (-2 \times -3) + (0 \times 0) + (4 \times 0) + (6 \times -4) = -18$$

$$Y_{2,1} = (-2 \times 0) + (0 \times 6) + (4 \times -1) + (6 \times -3) = -22$$

$$Y_{2,2} = (-2 \times 6) + (0 \times 0) + (4 \times -3) + (6 \times 0) = -24$$

$$Y_{2,3} = (-2 \times 0) + (0 \times -4) + (4 \times 0) + (6 \times 2) = 12$$

$$Y_{3,1} = (-2 \times -1) + (0 \times -3) + (4 \times 5) + (6 \times 0) = 22$$

$$Y_{3,2} = (-2 \times -3) + (0 \times 0) + (4 \times 0) + (6 \times 0) = 6$$

$$Y_{3,3} = (-2 \times 0) + (0 \times 2) + (4 \times 0) + (6 \times 3) = 18$$

The resulting feature map is:

$$\begin{bmatrix} 32 & 14 & -18 \\ -22 & -24 & 12 \\ 22 & 6 & 18 \end{bmatrix}$$

10.6)

Solution:

10.7)

Solution:

10.8)

Solution: